

Matthew Sebastian Kurniawan

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PROFILE

Final-year Information Systems and Technology student at Bandung Institute of Technology with a minor in Data Science and Artificial Intelligence, focused on taking AI systems from experiment to production. Currently researching Indonesian speech AI at Indosat Ooredoo Hutchison, with prior work shipping AI-assisted document processing into a live SaaS product on AWS. Interested in AI Engineering, AI Research, and Software Engineering roles.

WORK EXPERIENCE

INDOSAT OOREDOO HUTCHISON

June 2026 - Present

AI Researcher - Internship

- Fine-tuned Whisper large-v3 with LoRA (PEFT) on Indonesian call center audio, running a cumulative data-scaling experiment to quantify how transcript quality improves with data volume, then converted the merged model to CTranslate2 (faster-whisper) for production inference.
- Researched and evaluated real-time speech-to-speech frameworks for orchestrating STT, LLM, and TTS, then extended the selected framework with a new STT backend written from scratch (inference handler, argument class, backend registration) so models could be swapped through a single CLI flag.
- Replaced all three default components, none of which supported Indonesian, with a LoRA fine-tuned Whisper for STT, a fine-tuned Qwen LLM, and a voice-cloned TTS, then ran them as a real-time VAD-STT-LLM-TTS pipeline behind the OpenAI Realtime API and tuned turn-taking for noisy call center audio, reaching 0.504 s STT latency and under 2 s per full turn.
- Benchmarked MERaLiON, emotion2vec, and SpeechBrain on 700 labeled clips, then raised UAR from 0.822 to 0.868 by remapping output classes with no additional training, the project's largest gain.

ORGANIC STEEL GROUP

Apr 2026 - Present

Software Developer - Freelance

- Built AI-assisted document processing (OCR) that extracts structured data from invoices and receipts, including vendor, date, VAT breakdown, and line-item detail, removing manual entry for these fields.
- Developed and maintained a cloud-based SaaS application end-to-end using Next.js, React, Node.js/NestJS, Prisma ORM, and PostgreSQL, serving 30+ users.
- Delivered Stripe payment integration and multilingual support across 9 languages, and owned deployment and production maintenance on AWS (Elastic Beanstalk, RDS, S3, Amplify, CloudFront).

ASA KREASI INTERASIA

Jan 2026 - Apr 2026

Developer - Internship

- Contributed to a web-based ERP system using Java Spring Boot, implementing application features and maintaining system functionality against project requirements.
- Translated user and business requirements into technical implementations while collaborating with the team to deliver development tasks on schedule.

ASTRA HONDA MOTOR

Aug 2025 - Sep 2025

Data Analyst - Internship

- Built Python notebook workflows to compute sampling ratios and select transactions from Excel reports containing 1,000+ line items, replacing a manual review that took auditors 1 full working day per report.
- Automated Excel report generation classifying sampled and non-sampled transactions, producing audit-ready output consistently and removing manual formatting work.

COMPUTATIONAL THINKING LAB

Oct 2024 - Dec 2024

Practicum Assistant

- Assisted and supervised 79 students during Python programming practicum sessions, and evaluated coding assignments on correctness and code readability.

UNIVERSITY PROJECTS

Fraud Detection Machine Learning Models (Python)

Dec 2025

- Implemented Decision Tree and Logistic Regression from scratch, without ML libraries, on a large-scale synthetic dataset, covering preprocessing, feature engineering, model training, and validation analysis, placing 4th on the leaderboard with a ROC-AUC of 0.620.

ChemLab – Interactive Learning Platform (React Native, TypeScript, Supabase)

Jan 2026

- Built a cross-platform educational application supporting interactive chemistry learning through structured learning pathways, quizzes, virtual laboratory simulation, and an authentication system with Google OAuth, real-time database subscriptions, and progress tracking.

KAI Ticketing and Train Operation Database Modeling (MySQL)

Nov 2024

- Designed a relational database schema for railway ticketing and operational management by modeling entity relationships, performing normalization, and supporting operational workflow representation for Kereta Api Indonesia.

EDUCATION

BANDUNG INSTITUTE OF TECHNOLOGY

Aug 2023 - Present

Bachelor of Information Systems and Technology | Minor in Data Science & Artificial Intelligence | GPA: 3.54 / 4.00

- Coursework in Artificial Intelligence, Machine Learning, Data Mining, and Strategic Algorithms, focusing on pattern discovery, optimization, and data-driven problem solving.
- Strong foundations in Algorithms and Data Structures (C) and Object-Oriented Programming (Java), plus end-to-end web application development covering system design, requirement analysis, implementation, and testing.

COURSES & CERTIFICATIONS

UNIVERSITY OF TOKYO – ONLINE (JAPAN)

Sep 2025 - Dec 2025

Global Consumer Intelligence (GCI) Course 2025

- Completed the data science course run by Matsuo-Iwasawa Laboratory, Graduate School of Engineering, covering machine learning fundamentals, data analysis workflows, and practical AI modeling, and passed the final assessment.

ACHIEVEMENTS

- 3rd Place, Informatics Festival Hackathon – Padjadjaran University, 2025
- Finalist, Data Slayer 3.0 (National Data Science Competition) – Telkom University, 2025
- Finalist, PMJ Mathematics Competition, 2021
- Participant, BRAND UI Business Case Competition – University of Indonesia, 2024

EXPERIENCES (ORGANIZATIONAL/VOLUNTEERING)

HIMPUNAN MAHASISWA INFORMATIKA ITB

Aug 2025 - Dec 2025

Academic Division Staff

- Collected and organized learning resources and academic notes to support learning accessibility for fellow students within the cohort

ADDITIONAL

Technical Skills: AI Engineering, Machine Learning, Speech AI (ASR, speech emotion recognition), Computer Vision, Model Fine-tuning & Evaluation, Data Analysis, Web Development, Cloud Computing

Programming Languages: Python, Java, C, JavaScript, TypeScript, SQL, Assembly

AI/ML Stack: PyTorch, HuggingFace Transformers, PEFT/LoRA, NVIDIA NeMo, ONNX Runtime, CTranslate2 / faster-whisper, Whisper, Qwen, pyannote, Silero VAD, OpenAI Realtime API

Frameworks & Technologies: Next.js, React, React Native, Node.js/NestJS, Express.js, Spring Boot, FastAPI, WebSocket, Prisma ORM, Supabase, MySQL, PostgreSQL, AWS, Git

Languages: Indonesian (native), English (professional working proficiency)